

AI Health Screening Tool: Validation Results Report

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Overview

This report documents the validation work conducted on our AI-powered health screening tool prior to and during deployment in a randomized controlled trial in rural Rajasthan, India. The tool uses a large language model (LLM) to conduct structured health screenings through trained field staff, generating severity scores and health summaries for patients aged 40+ in underserved communities.

Our validation framework comprises four categories of evidence: (1) simulated clinical vignette testing against established medical benchmarks, (2) test-retest reliability analysis of severity score consistency, (3) medical expert review of real screening transcripts from the field, and (4) pilot field evidence on care-seeking behavior. We also describe validation phases that are in progress or planned.

This is a living document. Results will be updated as additional validation data become available.

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1. Clinical Vignette Simulations

1.1 Methodology

We developed an automated vignette simulation framework to benchmark our AI screening tool against clinical standards. The framework orchestrates conversations between an LLM-powered “Doctor Simulator” using our screening prompt and a scripted “Patient Simulator” that responds from standardized case definitions.

Case definitions. We adapted three clinical vignettes from Das et al. (2016, *American Economic Review*) — asthma, dysentery, and unstable angina — which were originally developed for standardized patient (SP) assessments of healthcare provider quality in rural India. The use of SP-based vignettes for measuring clinical competence was established by Das & Hammer (2005). We supplemented these with two additional vignettes (pre-eclampsia and tuberculosis) to broaden coverage of conditions relevant to the target population. The five conditions span common and serious presentations. Each case includes a structured set of history and examination questions with predefined correct answers, an answer key specifying the correct diagnosis, and a list of correct, palliative, and unnecessary/harmful treatments where an answer key is available.

Simulation procedure. The Doctor Simulator conducts the clinical interview following our screening prompt, dynamically asking follow-up questions based on patient responses. The Patient Simulator uses semantic similarity matching to map the doctor’s free-text questions to the closest predefined question-answer pair, ensuring consistent patient information across simulations. Each simulation produces a full conversation transcript, an extracted diagnosis, and treatment recommendations.

Evaluation metrics. Simulated consultations were scored on three dimensions:

- **Checklist completion rate:** The proportion of clinically recommended questions asked, following the methodology of Das et al. (2016). These checklists define essential history-taking and examination items for each condition.
- **Diagnostic accuracy:** Whether the LLM’s extracted diagnosis matched the correct diagnosis from the answer key.
- **Treatment appropriateness:** Whether recommended treatments were classified as correct, palliative, neutral, missing, or unnecessary/harmful based on the answer key and structured clinical review.

Scale. This analysis focuses on the adult-patient vignette profiles relevant for the AI screening tool: tuberculosis, pre-eclampsia, unstable angina, and asthma. Each was run three times, for 12 completed adult vignette simulations. The child dysentery vignette is excluded from the headline analysis because the tool is designed to screen adults, and that profile presents a father speaking on behalf of a two-year-old child. The pre-eclampsia vignette is included as an adult-patient safety check, not as a target-age field case.

1.2 Results

Overall performance. Across the adult vignette simulations, the AI screening tool achieved:

- **94% general screening checklist completion** (proportion of general screening items asked)
- **100% correct diagnosis rate** (top-1 diagnosis matched the answer key in all 12 adult vignette runs)
- **100% of benchmark adult runs with at least one correct treatment recommendation**

For the two adult Das benchmark cases with detailed case-specific checklist answer keys (unsta-

ble angina and asthma), the app completed **68%** of recommended vignette checklist items. This is lower than the general screening checklist completion because emergency presentations can appropriately trigger rapid escalation rather than exhaustive history-taking. Treatment recommendations across all 12 adult vignette runs included 51 correct items, 45 palliative items, 6 neutral items, 1 unnecessary or harmful item, and no missing treatment reviews.

Comparison with real-world providers. Das et al. (2016) measured the quality of care delivered by actual healthcare providers in rural and urban India using standardized patient (SP) vignettes, finding that providers completed an average of 27% of recommended checklist items and achieved correct diagnoses in only 6% of cases. Das et al. (2022) document similarly low quality across primary care markets in rural Indian villages. Our AI tool substantially outperforms these benchmarks on information gathering and diagnostic accuracy.

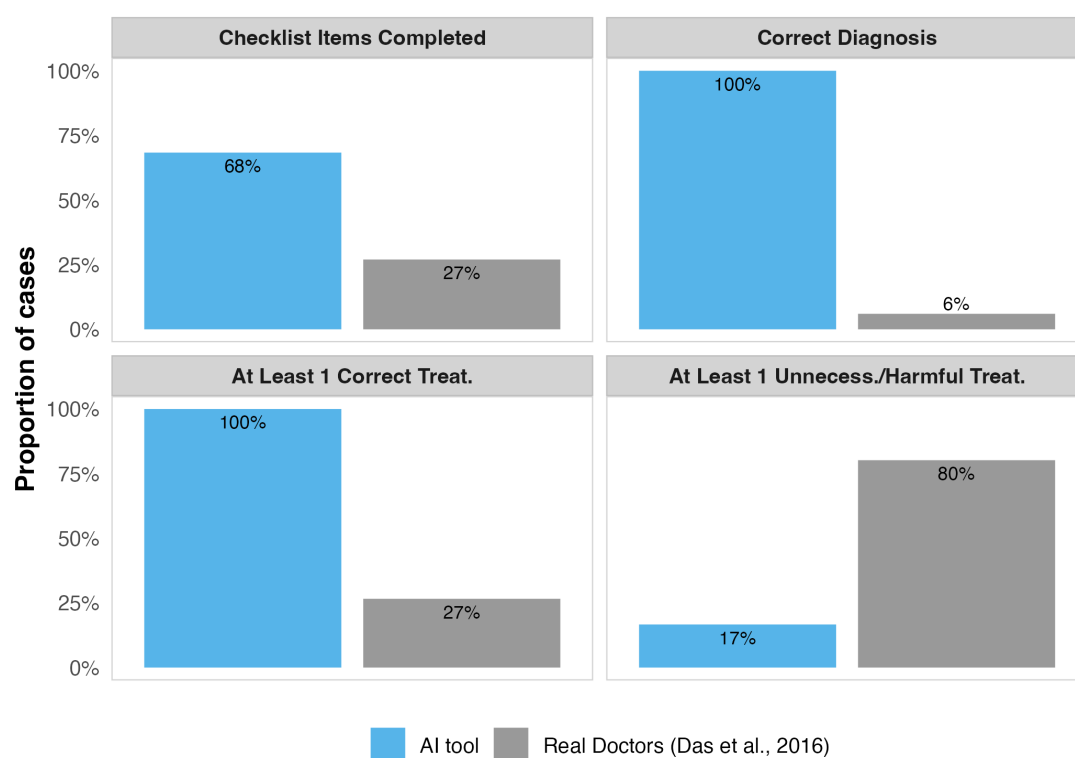


Figure 1. Comparison of AI model performance against real-world provider benchmarks from Das et al. (2016). The AI tool achieves substantially higher checklist completion and diagnostic accuracy, and substantially fewer unnecessary or harmful treatment recommendations.

1.3 Limitations

- Vignette patients respond from a fixed script; real patients provide less structured information.
- The semantic matching approach may overstate the AI's ability to interpret natural conversational responses.

- Cases are drawn from conditions well-represented in medical literature; performance on rarer or more ambiguous presentations has not been tested in simulation.
- The pre-eclampsia vignette is a useful safety check, but it is not a target-age field case for a study focused on adults aged 40+.

2. Test-Retest Reliability

2.1 Methodology

To assess the consistency of the AI tool’s severity scoring, we conducted a fixed-transcript test-retest reliability analysis. We selected 10 complete screening transcripts from field deployment in rural Rajasthan and submitted each transcript through the summary pipeline 10 times, generating 100 independent severity score assessments.

The AI tool generates a **severity score** on a 0–10 scale for each patient (“How important is it that this patient see a healthcare provider?”). This is the primary score used in the intervention to determine whether patients receive health information and referral recommendations.

This fixed-transcript design holds the clinical information constant and reruns only the summary step. It is therefore the fairest reliability test for the AI tool’s severity score: it isolates summary variation from variation in the live conversation path. Prior AI outputs were excluded from the input, so each run generated a fresh score from the transcript alone. All 100 fixed-transcript runs completed successfully.

2.2 Results

Severity scores are highly stable when the same transcript is scored repeatedly. Across all 10 transcripts, every repeated score was within ± 1 point of the same transcript’s own mean score. Four of the 10 transcripts received exactly the same score in all 10 repetitions, and every transcript had a maximum score range of 1 point or less. The mean within-transcript standard deviation is 0.25 points on the 10-point scale, confirming that stochastic variation across repetitions is small relative to the between-patient variation that drives clinical decisions.

Metric	Result
Completed summaries	100 / 100
Pairwise run-to-run comparisons within ± 1 point	100%
Pairwise exact run-to-run agreement	79%
Transcripts with maximum range ≤ 1 point	10 / 10
Transcripts with identical score in all 10 repetitions	4 / 10
Mean within-transcript standard deviation	0.25 points
Mean within-transcript range	0.60 points

This section measures repeatability of the app against itself. Agreement with medical expert benchmarks is reported separately in Section 3.

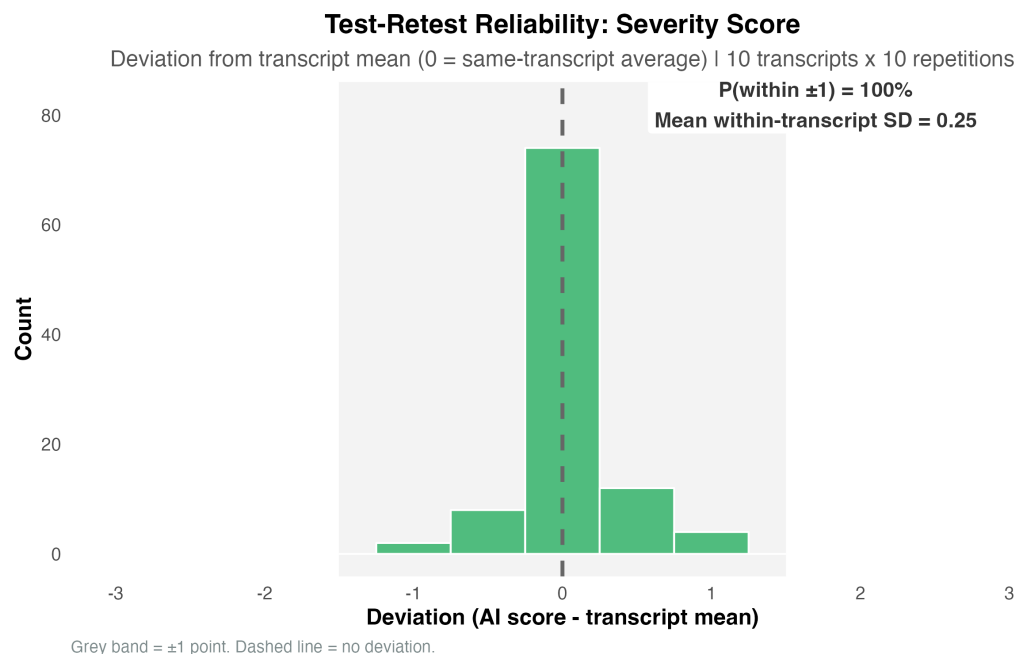


Figure 2. Distribution of fixed-transcript severity score deviations from each transcript's own mean score. All regenerated scores fall within ± 1 of the same-transcript mean. Grey band = ± 1 point.

2.3 Limitations

- The sample is limited to 10 field transcripts from the expert-review process.
- Fixed-transcript reliability tests score stability conditional on the same transcript; it does not test whether repeated live conversations would ask exactly the same follow-up questions.
- The exercise tests severity-score stability, not whether the transcript itself captured all clinically relevant information.

3. Phase 1 Expert Review

3.1 Methodology

Four senior medical experts independently reviewed the AI screening prompt and 10 curated transcripts from field screening in rural Rajasthan. The expert panel was assembled by co-PI Dr. Mona Duggal (ICMR-NIRDHDS) and includes specialists in community medicine and maternal/child health, public health administration and digital health (including India's National NCD Portal), clinical practice in the study area, and Indian Public Health Standards (IPHS) and National Health Mission (NHM) implementation.

Each expert was asked to: (1) write a structured report evaluating the prompt's medical soundness,

(2) provide annotated feedback on each transcript with severity ratings, and (3) participate in a debrief call with the research team. This phase was unblinded — experts knew the outputs were AI-generated — and was designed to assess medical soundness, severity calibration, and operational risks before scaling the intervention.

3.2 Severity Score Agreement

Experts independently rated each transcript on a 0–10 severity scale (“How important is it that this patient see a healthcare provider?”). We compare the AI tool’s fixed-transcript severity scores against the mean expert rating for each transcript.

The AI tool is largely unbiased relative to expert consensus, with a mean bias of -0.26 points (AI minus mean expert). These are validity metrics against expert benchmarks, separate from the test-retest metrics in Section 2. Across 100 fixed-transcript summary runs, severity scores fall within ± 1 point of the mean expert score 78% of the time. At the transcript level, the mean AI score across 10 repetitions is within ± 1 point of the expert benchmark for 7 of 10 transcripts.

Comparison	Pearson r	n
AI tool mean vs. expert mean	0.93	10 transcripts
Expert vs. expert	0.81	28 pairs

The AI tool’s transcript-level correlation with expert consensus is high in this small benchmark set. Three of four experts show very high agreement with each other and with the AI. One expert systematically rates acute cases (chest pain, breathlessness) as moderate where the other experts and the AI rate them as urgent, reflecting a clinical judgment difference about over-triage risk in resource-constrained settings. Figure 3 shows the distribution of AI bias across transcripts: the AI tends to rate slightly below the expert mean, but the average bias is small.

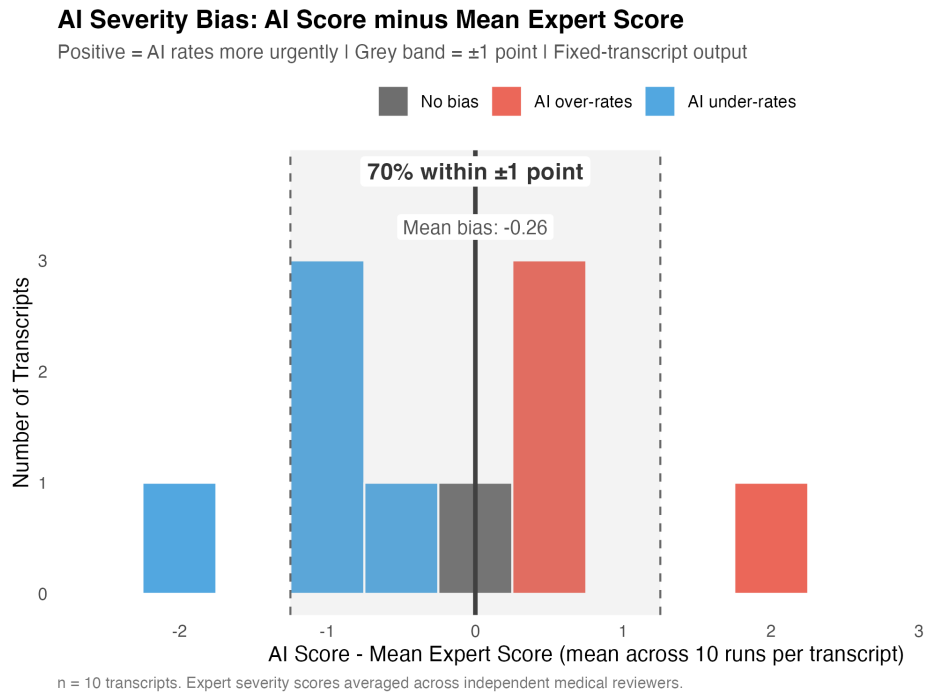


Figure 3. Distribution of AI severity bias (AI score minus mean expert score) across 10 transcripts. AI score is the mean of 10 fixed-transcript runs for each transcript. Mean expert score is the average severity rating across independent medical reviewers.

3.3 Key Qualitative Findings

Areas of improvement identified by experts

1. The AI should avoid asking multiple questions simultaneously, because elderly patients often respond only to the final question.
2. The AI should ask more about current medications, prior diagnoses, and previous care-seeking.
3. Referral recommendations must be realistic for rural healthcare infrastructure and aligned with Indian Public Health Standards facility tiers.

The AI tool emphasizes these priorities: acute presentations such as chest pain trigger urgent escalation, questions are asked one at a time, symptom history is probed systematically, and referral guidance is framed around feasible provider visits rather than detailed treatment instructions.

3.4 Overall Expert Verdicts

- “Highly medically sound for a community-level intervention.” Confidence: *high, conditional on safety guardrails.*
- “The AI tool carries huge potential in primary screening and referral.” Confidence: *high, conditional on interviewer training.*

- “The potential is high... if medical history and medications are well recorded, the AI would be successful.” Confidence: *high, conditional on deeper probing*.
 - One expert found the tool promising but limited by potential over-diagnosis tendencies. Confidence: *limited*.
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4. Pilot Field Evidence

Sections 1–3 evaluate the AI screening tool’s clinical validity through simulations, reliability testing, and expert review. This section presents complementary evidence from the tool’s pilot field deployment, addressing whether providing screening results encourages patients to seek care.

4.1 Pilot Design and Sample

We conducted a pilot RCT in villages in Udaipur district, Rajasthan, in March–May 2026. Trained field surveyors administered the AI screening tool to adults aged 40+. All participants received the screening, but only treatment households received the intervention: a printed summary of the AI’s health assessment, a guided care-seeking planning exercise, and a brief session with a household decision-maker where feasible to discuss the screening results and plan a provider visit together.

The eligible baseline analysis sample is the screen-positive sample with severity score ≥ 4 , excluding self-selected respondents; this includes 275 respondents. We found 246 respondents for the endline survey, after an average of 18.7 days. Provider-visit outcomes are observed for 244 respondents in the matched panel. The endline survey measured provider visits and facility type after the baseline screening.

For context, Figure 4 shows the baseline severity distribution for all baseline respondents with a nonmissing severity score, including respondents who are not in the matched panel and those below the score ≥ 4 analysis threshold. Baseline severity in this full baseline sample is centered around 6.6, with scores ranging from 0 to 9.

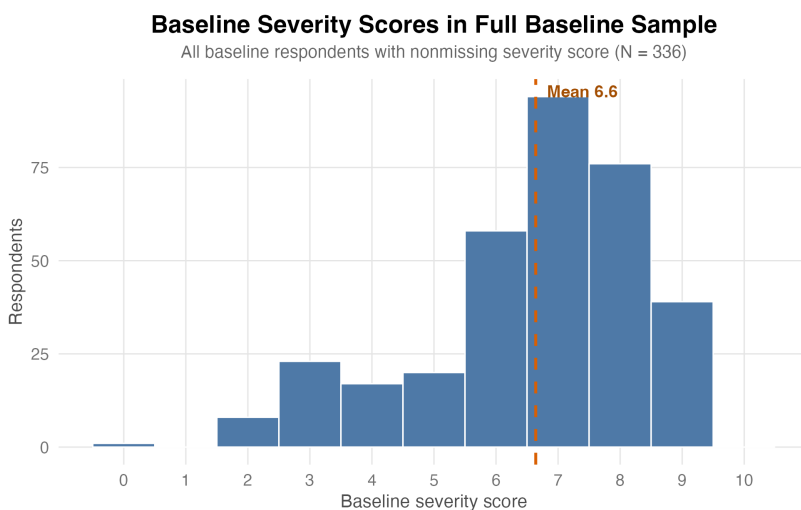


Figure 4. Baseline severity-score distribution among all baseline respondents with a nonmissing severity score (N = 336), including unmatched respondents and respondents below the score ≥ 4 analysis threshold.

4.2 Effect of Screening Information on Care-Seeking

We estimate the effect of the treatment bundle — receiving the AI screening results, completing a guided care-seeking planning exercise, and involving a household decision-maker — on whether respondents visited a healthcare provider after the screening. Figure 5 shows three outcomes: (1) any provider visit in the full matched sample, (2) any provider visit among respondents who had not visited a provider in the 30 days before baseline, and (3) CHC visits in the full matched sample. In the rural Indian public health system, a Community Health Centre (CHC) is a higher-capacity public facility than a sub-centre or primary health centre: it is meant to provide more comprehensive, higher-quality clinical care, and care is free in principle.

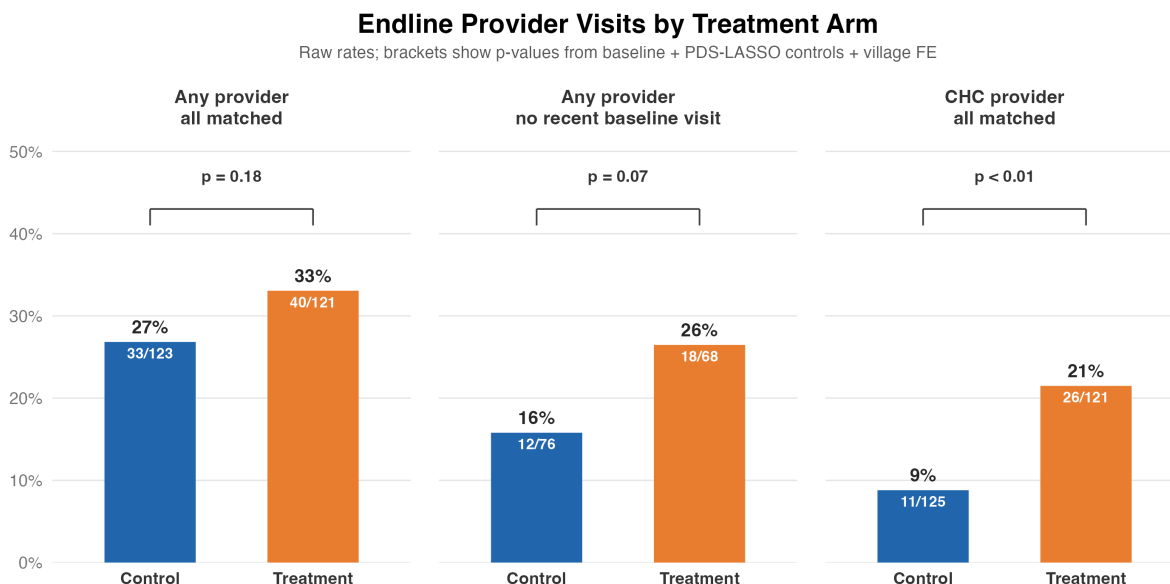


Figure 5. Proportion of respondents who visited a healthcare provider after baseline screening, by treatment status. Bars show raw rates. Bracket labels report p-values from the baseline plus PDS-LASSO control specification with village fixed effects.

Note: The p-values shown in Figure 5 come from the baseline-control plus post-double-selection LASSO specification, using demographic controls selected by LASSO, village fixed effects, and household-clustered robust standard errors. These are exploratory pilot estimates, not definitive impact estimates. Village-clustered p-values are not used here because the pilot has too few true village clusters for reliable village-level inference.

The treatment arm shows positive but noisy movement on any provider visit in the full matched sample: 33% (40 / 121) of treatment respondents visited a provider, compared with 27% (33 / 123) in the control group. The adjusted treatment effect is 8.3 percentage points, with a descriptive p-value of 0.18.

The effect is larger among respondents who had not visited a provider in the 30 days before baseline. In that group, 26% (18 / 68) of treatment respondents visited a provider after baseline, compared with 16% (12 / 76) of control respondents. The adjusted treatment effect is 13.5 percentage points, with a descriptive p-value of 0.07. Since this is the group that needs the intervention most, we will update the main RCT to screen based on this outcome or use it as pre-specified heterogeneity.

The largest point estimate is for CHC visits. In the full matched sample, 21% (26 / 121) of treatment respondents visited a CHC after baseline, compared with 9% (11 / 125) of control respondents. The adjusted treatment effect is 13.7 percentage points, with a descriptive p-value below 0.01. The large increase in CHC visits suggests that the quality of healthcare accessed may be improving as well.

4.3 Interpretation

The pilot is not powered as a definitive impact evaluation, but the direction of the evidence is encouraging. The treatment bundle appears to shift provider visits modestly in the full sample, more among respondents without a recent provider visit, and especially toward CHC visits. The larger movement among recent non-visitors motivates using recent non-visit status as a screening or targeting criterion in the main study, so the intervention is concentrated among people whose baseline behavior leaves more scope for a care-seeking response.

5. Planned and In-Progress Validation

5.1 Blinded Transcript Validation

Design. Medical experts independently review screening transcripts without seeing the AI's outputs, providing their own severity assessments for comparison. Unlike Phase 1, this phase is blinded and uses a larger, representative sample of transcripts.

Outcome measures:

- Agreement rate between AI and expert severity scores (within ± 1 point)
- False negative rate: proportion of high-severity cases (expert score ≥ 7) rated low by AI (score ≤ 4)
- False positive rate: proportion of low-severity cases (expert score ≤ 3) rated high by AI (score ≥ 7)
- Summary quality scores

Acceptance criteria:

- Agreement rate $\geq 80\%$
- False negative rate $\leq 5\%$ (critical safety threshold)
- False positive rate $\leq 20\%$

Following any significant prompt modifications, a verification round of additional transcripts will confirm improvements.

5.2 Live Field Observation

Design. Medical experts will accompany field teams to observe live screening conversations as they occur naturally. For each interaction:

1. The expert observes silently during the AI-mediated screening, noting conversation flow, questions asked, and patient responses.
2. After the screening, the expert conducts a brief independent clinical assessment of the patient.
3. Expert findings are compared with AI-generated outputs to evaluate real-world performance.

Assessment criteria: Completeness of symptom elicitation, accuracy of patient response interpretation, severity score calibration against clinical judgment, recommendation actionability, and whether any concerning symptoms were missed.

5.3 Red Teaming

Research assistants and medical consultants will attempt to elicit failures from the AI tool in targeted scenarios: ambiguous or conflicting symptom presentations; acute emergency presentations where appropriate escalation is critical; boundary cases where the tool should decline to provide diagnoses or treatment recommendations; culturally sensitive topics; contradictory or incomplete patient responses; and Hindi comprehension across regional dialect variations. The exercise will produce a documented log of identified failure modes and corresponding prompt fixes, with re-testing to confirm that critical issues have been resolved.

5.4 Ongoing Monitoring During Main RCT

Throughout the main study, continuous monitoring will be maintained:

- **Weekly transcript review:** randomly selected transcripts reviewed by the ICMR-NIRDHDS medical expert panel.
- **Escalation protocol:** flagged transcripts trigger immediate senior expert review, participant follow-up if clinically indicated, and root cause analysis.
- **Algorithmic fairness audits:** regular checks on AI performance across gender, age, caste, and baseline care-seeking status to detect systematic bias.

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